SynDiff Synthetic Benchmark — Live Progress

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478/942 jobs complete (50.7%)

S = I(X_1;Y|X_2) − I(X_1;Y) in nats; S>0 synergy-dominant, S<0 redundancy-dominant. SynDiff (ours) vs prior estimators.

Dataset construction

Phase 1 — Gaussian (closed-form ground truth)

Scalar base: $X_1, X_2 \sim \mathcal{N}(0,1)$ with $\mathrm{corr}(X_1, X_2) = \rho$; label $Y = X_1 + X_2 + \sigma_y\varepsilon$, $\varepsilon \sim \mathcal{N}(0,1)$. Thus $\mathrm{Var}(Y) = 2 + 2\rho + \sigma_y^2$ and $\mathrm{Cov}(X_i, Y) = 1 + \rho$. To lift to d dimensions the informative scalar is padded with $d-1$ i.i.d. $\mathcal{N}(0,1)$ coordinates and rotated by a fixed per-seed orthogonal matrix Q_i (one per variable). Rotation and independent padding preserve mutual information, so S is unchanged and the closed form carries over. Ground truth is computed from the 3×3 covariance via $I = \frac{1}{2}\log\big(\det\Sigma_A \det\Sigma_B / \det\Sigma_{AB}\big)$ and the analogous conditional form. Cells sweep $\rho \in \{-.9, \dots, .9\}$, $\sigma_y \in \{.1, 1\}$, $n \in \{1\text{k}, \dots, 100\text{k}\}$, $d \in \{1, 8, 32, 128\}$.

Phase 2 — discrete U/R/S (mixture-density MC truth)

Latent sources $s_1, s_2, s_3 \sim \mathrm{Unif}\{0, \dots, Q-1\}$ with $Q = 8$. Encodings $X_i = W_i\,\mathrm{onehot}(s_i) + \sigma_{\mathrm{enc}}\,\mathcal{N}(0, I_{32})$ with orthonormal $W_i \in \mathbb{R}^{32\times 8}$ frozen per seed. Regimes (X_1 always encodes s_1): **UNIQUE** $Y{=}s_1$; **REDUNDANT** $Y{=}s_1$ and X_2 also encodes s_1; **SYNERGY1** $Y{=}(s_1{+}s_2) \bmod Q$ with $X_2 \sim s_2$; **SYNERGY2** $Y{=}(s_3{+}(s_1{+}s_2) \bmod Q) \bmod Q$ with $X_2{=}[\mathrm{enc}(s_2)\,|\,\mathrm{enc}(s_3)] \in \mathbb{R}^{64}$; **INDEPENDENT** $Y{=}s_3$. The λ-dial mixes the SYNERGY1 and REDUNDANT rules per sample (Bernoulli(λ)). Ground truth is the exact $S = I(X_1; X_2, Y) - I(X_1; X_2) - I(X_1; Y)$ obtained by Monte Carlo over the enumerable Gaussian-mixture components (matches the discrete limit as $\sigma_{\mathrm{enc}} \to 0$: $\pm\log Q$ for pure synergy/redundancy, 0 for unique/independent).

Method: SynDiff

Target $S = I(X_1; Y \mid X_2) - I(X_1; Y)$ in nats. By the symmetry of interaction information this equals $I(X_1; X_2 \mid Y) - I(X_1; X_2)$, which the four-mask form below estimates directly. **VP schedule:** $\beta(t) = 0.1 + 19.9t$, $B(t) = \int_0^t \beta = 0.1t + 9.95t^2$, signal $k(t) = e^{-B/2}$, noise $\sigma^2(t) = 1 - e^{-B}$, weight $w(t) = \beta(t)/(2\sigma^2(t))$. A single noise-prediction network $\hat\varepsilon(x_{1,t}, t, \mathrm{cond})$ (*MaskedCondMLP*: Fourier-t features ⊕ per-slot conditioning embeddings each with a learned NULL token; 4×512 SiLU) is trained to denoise X_1 while each conditioning slot (X_2, Y) is independently dropped to its NULL token with probability 0.15. EMA weights (decay 0.999) are used at evaluation. **Four-mask estimator.** Per evaluation triple and per stratified time t (K=16 strata on $[10^{-3}, 1]$) with *one shared* noise draw ε and $x_{1,t} = k(t)x_1 + \sigma(t)\varepsilon$:

$$
\begin{aligned}
e_1 &= \hat\varepsilon(x_{1,t}, t, x_2, y), & e_2 &= \hat\varepsilon(x_{1,t}, t, \varnothing, y),\\
e_3 &= \hat\varepsilon(x_{1,t}, t, x_2, \varnothing), & e_4 &= \hat\varepsilon(x_{1,t}, t, \varnothing, \varnothing),\\
\mathrm{integrand}(t) &= w(t)\big(\|e_1 - e_2\|^2 - \|e_3 - e_4\|^2\big).
\end{aligned}
$$

Per-sample $\hat S = (1 - 10^{-3})\,\overline{\mathrm{integrand}}_t$; the estimate is the mean over 10^4 fresh triples with a 1000-sample bootstrap CI (norms in fp32, no-grad, EMA weights). **Baselines:** *minde_comp* (three separately trained MINDE MI terms composed as above), *MINE* and *InfoNCE* (lower-bound MI, same composition), *batch_pid* (Liang et al. CVX PID, k-means k=20 then max-entropy solve), and *broja/plugin* on d=1.

Sweep progress

Phase	Method	Done	Expected
P1	batch_pid	30	96
P1	broja	0	6
P1	infonce	96	96
P1	minde_comp	96	96
P1	mine	96	96
P1	plugin	2	6
P1	syndiff	96	96
P2	batch_pid	1	90
P2	infonce	15	90
P2	minde_comp	13	90
P2	mine	16	90
P2	syndiff	17	90

Accuracy vs ground truth (completed cells, lower error better)

Method	#cells	mean |est−truth|	sign acc
minde_comp	37	0.0489	0.89
plugin	2	0.1343	1.00
syndiff	38	**0.1414**	0.92
batch_pid	31	0.5450	0.71
mine	38	0.7592	0.66
infonce	37	4.9664	0.43

SynDiff vs prior — head to head (nats)

Cell	truth	**SynDiff**	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rho0_sy0p1_d8_n20k	+1.963	**+1.887**	+0.147	+1.922	+0.631	-5.604	–
p1_dial_rho0_sy1_d8_n20k	+0.144	**+0.136**	+0.154	+0.145	-0.555	-5.315	–
p1_dial_rho0p3_sy0p1_d8_n20k	+1.740	**+1.677**	+0.128	+1.678	+0.627	-5.460	–
p1_dial_rho0p3_sy1_d8_n20k	+0.007	**+0.006**	+0.129	+0.010	-0.603	-5.429	–
p1_dial_rho0p5_sy0p1_d8_n20k	+1.477	**+1.407**	+0.115	+1.434	+0.525	-5.436	–
p1_dial_rho0p5_sy1_d8_n20k	-0.134	**-0.134**	+0.108	-0.128	-0.829	-5.254	–
p1_dial_rho0p7_sy0p1_d8_n20k	+1.035	**+0.971**	+0.090	+0.991	+0.309	-5.395	–
p1_dial_rho0p7_sy1_d8_n20k	-0.329	**-0.334**	+0.098	-0.321	-0.651	-5.190	–
p1_dial_rho0p9_sy0p1_d8_n20k	+0.024	**-0.032**	+0.055	-0.025	-0.329	-5.321	–
p1_dial_rho0p9_sy1_d8_n20k	-0.610	**-0.601**	+0.066	-0.612	-0.807	-5.145	–
p1_dial_rhon0p3_sy0p1_d8_n20k	+2.047	**+1.976**	+0.161	+2.017	+0.724	-5.385	–
p1_dial_rhon0p3_sy1_d8_n20k	+0.209	**+0.199**	+0.157	+0.210	-0.562	-5.363	–
p1_dial_rhon0p5_sy0p1_d8_n20k	+2.023	**+1.953**	+0.149	+1.980	+0.795	-5.507	–

Cell	truth	**SynDiff**	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rhon0p5_sy1_d8_n20k	+0.213	**+0.197**	+0.153	+0.214	-0.504	-5.342	–
p1_dial_rhon0p7_sy0p1_d8_n20k	+1.896	**+1.826**	+0.138	+1.853	+0.689	-5.488	–
p1_dial_rhon0p7_sy1_d8_n20k	+0.177	**+0.158**	+0.148	+0.182	-0.648	-5.374	–
p1_dial_rhon0p9_sy0p1_d8_n20k	+1.473	**+1.412**	+0.142	+1.445	+0.466	-5.470	–
p1_dial_rhon0p9_sy1_d8_n20k	+0.083	**+0.054**	+0.142	+0.089	-0.580	-5.200	–
p1_dim_rho0p7_sy1_d128_n20k	-0.329	**-0.050**	–	-0.295	-1.852	-6.133	–
p1_dim_rho0p7_sy1_d1_n20k	-0.329	**-0.329**	-0.240	-0.322	-0.321	-0.271	-0.191
p1_dim_rho0p7_sy1_d32_n20k	-0.329	**-0.365**	+0.133	-0.330	-1.467	-6.102	–
p1_dim_rhon0p7_sy1_d128_n20k	+0.177	**+0.022**	+0.136	+0.271	-1.938	-6.143	–
p1_dim_rhon0p7_sy1_d1_n20k	+0.177	**+0.168**	–	+0.184	+0.183	+0.171	+0.308
p1_dim_rhon0p7_sy1_d32_n20k	+0.177	**+0.129**	+0.137	+0.177	-1.085	-6.105	–
p1_ns_rho0p7_sy1_d8_n100k	-0.329	**-0.332**	+0.098	-0.321	-0.704	-5.204	–
p1_ns_rho0p7_sy1_d8_n1k	-0.329	**-0.331**	+0.098	-0.325	-0.630	-5.217	–
p1_ns_rho0p7_sy1_d8_n20k	-0.329	**-0.333**	+0.098	-0.322	-0.652	-5.261	–
p1_ns_rho0p7_sy1_d8_n5k	-0.329	**-0.331**	+0.098	-0.321	-0.651	-5.174	–
p1_ns_rhon0p7_sy1_d8_n100k	+0.177	**+0.157**	+0.149	+0.180	-0.648	-5.374	–
p1_ns_rhon0p7_sy1_d8_n1k	+0.177	**+0.160**	+0.148	+0.183	-0.497	-5.338	–
p1_ns_rhon0p7_sy1_d8_n20k	+0.177	**+0.154**	+0.148	+0.180	-0.602	-5.375	–
p1_ns_rhon0p7_sy1_d8_n5k	+0.177	**+0.154**	+0.148	+0.185	-0.453	-5.449	–
p2_redundant_senc0p1	-2.079	**-5.757**	–	-3.164	-2.058	-2.194	–
p2_redundant_senc0p5	-0.730	**-0.425**	–	-0.625	-0.924	-2.035	–
p2_redundant_senc1	-0.065	**-0.000**	–	–	-0.639	–	–
p2_unique_senc0p1	-0.000	**-0.000**	–	+0.009	-0.838	-2.042	–
p2_unique_senc0p5	-0.000	**+0.000**	–	+0.005	-1.128	-2.075	–
p2_unique_senc1	-0.000	**+0.000**	+0.148	+0.002	-1.070	-2.071	–